Strategic Price Setting

ECON 201: Principles of Microeconomics
Lecture 13

Div Bhagia

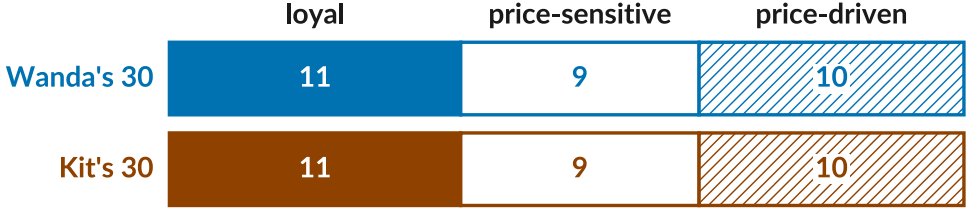
Last Class: Market Structures

- So far, a firm set its price taking its demand curve as given. It assumed its own price would have little effect on what competitors do.
- That is reasonable with many substitutes, each reaching a small part of the market, e.g., a city's many clothing shops.
- With only a few firms, one firm's price changes the demand the others face. E.g., three fast food places in a mall each know their prices affect the others.
- Pricing decisions are then interdependent, so each firm has to think strategically about what its rivals will do.

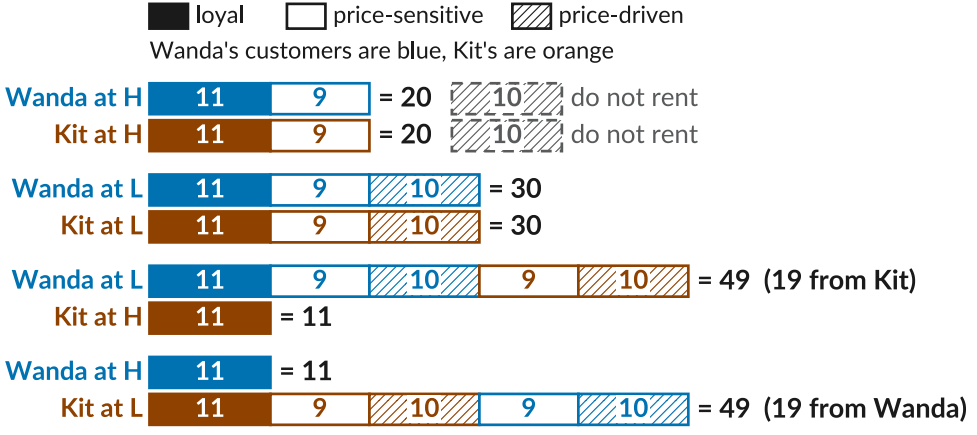
Today: how do firms set prices when each one has to think about what the others will do?

Wanda and Kit

- On a Southern California beach, Wanda rents windsurf boards and Kit rents kiteboards.
- Each rental costs its owner \$10, so MC $= 10$. Each picks a high price, $H = \$36$, or a low price, $L = \$20$, without knowing the other's.
- Each has 30 customers who prefer its sport. **Loyal** customers rent from it at any price, **price-sensitive** ones pay H but switch if the rival is cheaper, and **price-driven** ones pay L but not H.



Who Rents at Each Pair of Prices



The rival's price-sensitive and price-driven customers go to whichever firm is cheaper.
Try it: interactive version.

Game Theory

We can study this interaction using **game theory**, a set of models of strategic interactions.

- A **strategic interaction** is one where each person's outcome depends on what others do, and each person knows it.
- A **game** describes the players, what they can do (their **strategies**), and what each one gets (their **pay-offs**).
- Here, the players are Wanda and Kit, the strategies are *H* and *L*, and the pay-offs are their profits.
- It is a **simultaneous game**: each chooses without knowing what the other has chosen. They play only once.

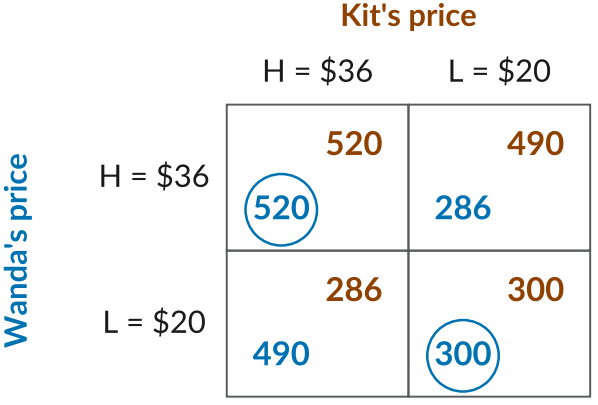
The Pay-off Matrix

- Profit is $(P - 10) \times$ customers.
- Both at H: $(36 - 10) \times 20 = 520$ each.
- Wanda at L, Kit at H: Wanda earns $(20 - 10) \times 49 = 490$, and Kit $(36 - 10) \times 11 = 286$.
- In each cell, Wanda's profit is at the bottom left and Kit's at the top right.

Wanda's price \ Kit's price	H = $36	L = $20
H = $36	520 / 520	490 / 286
L = $20	286 / 490	300 / 300

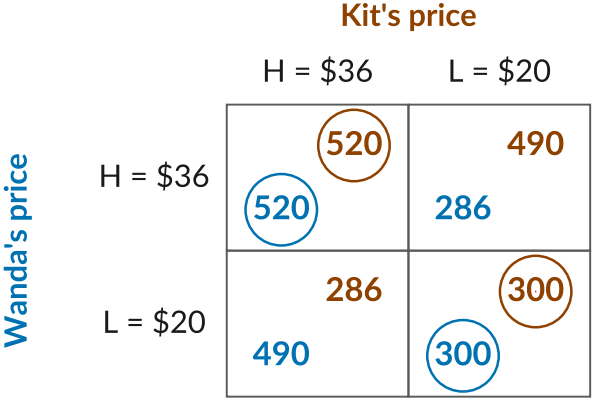
Best Response

- A **best response** is the strategy that gives a player the highest pay-off, given what the others choose.
- If Kit picks *H*: Wanda earns 520 at *H* and 490 at *L*, so she picks *H*.
- If Kit picks *L*: 286 at *H* and 300 at *L*, so she picks *L*.
- Blue rings mark her best responses. She matches Kit's price.

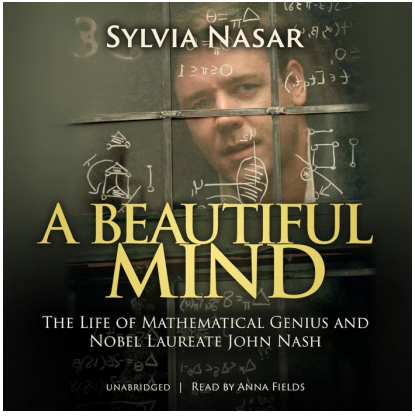


Nash Equilibrium

- A **Nash equilibrium**: each player's strategy is a best response to the others'.
- Orange rings mark Kit's best responses. (H, H) and (L, L) have both rings, so both are Nash equilibria.
- It is our prediction: neither firm wants to change on its own.
- Both prefer (H, H), but H is riskier. Which happens depends on what each expects.



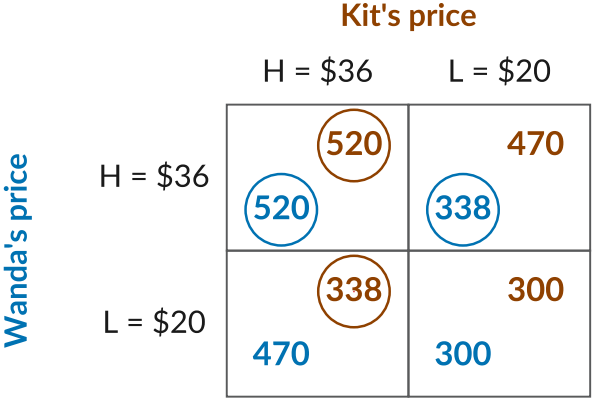
John Nash



- John Nash finished his 27-page doctoral thesis at Princeton at 21.
- It defined the Nash equilibrium, which transformed economics and won him a share of the 1994 Nobel Prize.
- His life is the subject of the book and film *A Beautiful Mind*.

More Loyal Customers

- Now each firm has 13 loyal and 7 price-sensitive customers.
- If Kit picks *L*, Wanda earns $26 \times 13 = 338$ at *H*, more than 300 at *L*.
- So *H* is her best response whatever Kit does: a **dominant strategy**. Same for Kit.
- More loyal customers: less elastic demand, and both charge high.



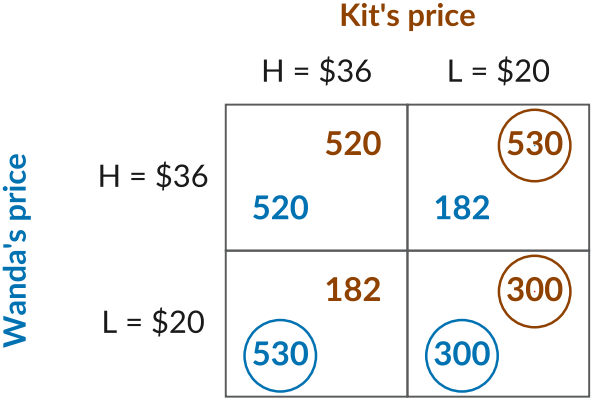
Fewer Loyal Customers

- Now each firm has only 7 loyal and 13 price-sensitive customers.
- At (H, L), the high-priced firm keeps just its 7 loyal customers: $26 \times 7 = 182$. The other gets 53: $10 \times 53 = 530$.
- What is each firm's best response? Where does the game end up?

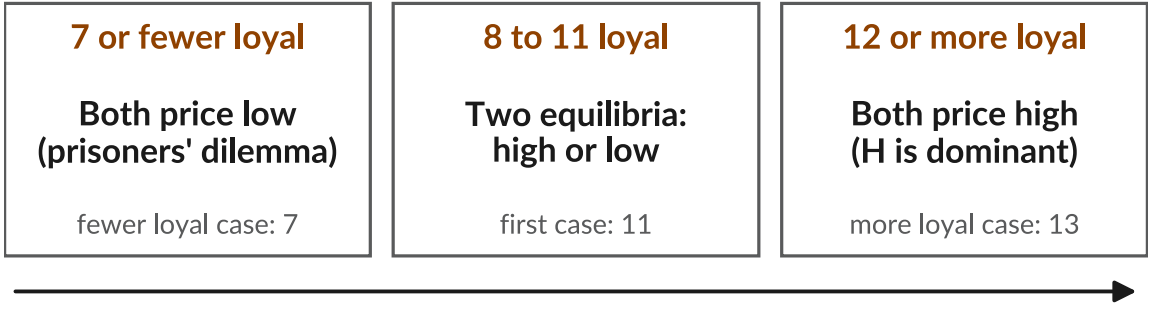
Wanda's price \ Kit's price	H = $36	L = $20
H = $36	520, 520	530, 182
L = $20	182, 530	300, 300

Prisoners' Dilemma

- *L* is dominant for both: 530 beats 520, and 300 beats 182.
- The only Nash equilibrium is (*L*, *L*), at 300 each, though both would earn 520 at (*H*, *H*).
- A **prisoners' dilemma**: each player has a dominant strategy, but another outcome gives every player more.
- Self-interest drives the price down, and customers gain.



Loyalty and Prices



- More loyal customers means the two boards are more differentiated, so demand is less elastic and prices are higher.
- Fewer loyal customers means buyers compare prices, so competition is more intense and prices are lower.

Try it: change the loyal customers in the interactive version.

What Wanda and Kit Tell Us

With only a few firms, each firm's best price depends on what its rivals charge. Two things follow:

1. **Differentiation still sets prices.** As with the markup rule, the more loyal the customers, the less elastic demand is and the higher the prices. When customers compare prices, competition pushes prices down.

2. **Self-interest can work against the firms and for the buyers.** Both would earn more charging high, but each gains by undercutting, so both end up low. Customers benefit, which is why few rivals are tempted to collude, and why cartels are illegal.

In our game they choose once. Firms that meet every day may each hold back from cutting for fear the other will match: one way few rivals keep prices high without ever agreeing to.

Midterm 1

- **Monday, October 12**, in class, 65 minutes. It covers Lectures 3 to 13.
- The exam has three parts, weighted equally: multiple-choice questions, short-answer problems, and one long-form question. Each part is its own sheet, so write your name and CWID on every sheet.
- The exam is worth 15% of your course grade. Closed book and closed notes.
- Bring a basic calculator. You will be given the help sheet.
- To prepare, follow the **study guide**. Then take the **practice exam** timed, with only the help sheet, and check it with the solutions.